

PREM KIRAN POLEPALLI

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EXPERIENCE

OPEN SOURCE WITH SLU (St Louis, MO, USA)

(January 2026 – Present)

Tech Lead, GradEval360 Project

- **Architected a greenfield Graduate Evaluation System** from the ground up using **React and Node.js**, translating complex university policy into a modular, role-based digital workflow.
- **Designed a secure data schema** in **PostgreSQL** to manage sensitive academic evaluations, implementing **Okta-based OIDC authentication** to ensure 100% compliance with FERPA-aligned security standards.
- **Reduced initial development overhead by 35%** by establishing a standardized component library with **Tailwind CSS**, enabling the team to build 10+ core functional modules within the first sprint cycle.
- Led weekly client demos with university stakeholders to refine UX requirements, managing a backlog of **50+ Jira tickets** and maintaining a **95% sprint completion rate**.

OPEN SOURCE WITH SLU (St Louis, MO, USA)

(August 2025 – December 2025)

Tech Lead, STL Data API Project

- Engineered the foundational architecture for a regional Open Data API, utilizing **Microservices** to aggregate disparate public transit and infrastructure datasets into a unified **RESTful gateway**.
- **Implemented a high-concurrency ingestion engine** using **Kafka and Node.js**, designed to process over **10,000+ simulated data points per second** with sub-50ms internal latency.
- Standardized API documentation using **Swagger/OpenAPI**, facilitating seamless integration for future frontend consumers and reducing integration friction for the 4-person development team.
- **Established a 100% automated CI/CD pipeline** via **GitHub Actions**, integrating automated linting and unit testing to ensure zero-regression during the initial build phase.

OPEN SOURCE WITH SLU (St Louis, MO, USA)

(May 2025 – June 2025)

Full Stack Developer, Mouser Project

- Integrated Rerum, a feature for storing scientist's annotations related to experiments, enabling efficient capture and management of textual insights.

CLARIVATE (Bangalore, India)

(August 2021 – July 2024)

Full Stack Developer

- **Optimized database performance** by migrating complex business logic from the backend to **Stored Procedures (T-SQL)**, reducing query execution time from **4 minutes to under 15 seconds** for high-volume data reports.
- Architected and maintained enterprise web applications using **ASP.NET Core and C#**, consistently maintaining 99.9% system uptime for global legal and IP clients.
- Reduced deployment cycles by 30% by re-engineering **Azure DevOps CI/CD pipelines**, automating unit testing via **MS Test** and reducing manual intervention.
- **Collaborated in an Agile environment** to deliver 20+ major feature releases, mentoring a team of 5 junior developers and ensuring 100% adherence to code quality standards.

AGNEYAS LABS (Bangalore, India)

(March 2021 - April 2021)

Software Intern

- Training in real-world operations of the Air quality website as well as a fundamental comprehension of the software used to show real-time data.

VERZEO (Bangalore, India)

(April 2020 - May 2020)

Business Analytics Intern

- Predicting the cost of used cars from the given data: To predict and understand the cost of used cars. The data is collected from various sources and distributed across various locations in India. The analysis was done on Jupyter Notebook platform using Python.

BHARAT SANCHAR NIGAM LIMITED (Anantapur, India)

(June 2019 - July 2019)

Computer Networks Intern

- Industrial training in the operation of wireless connections, the Internet, and broadband, as well as a fundamental understanding of the software needed to monitor the efficient operation of services.

EDUCATION

- MS in Computer Science

(August 2024 – May 2026)

Saint Louis University, Saint Louis, MO, USA | CGPA: 3.84 for 4

- Bachelor of Technology in Electronics and Communication Engineering

(July 2017 - June 2021)

Manipal Institute of Technology, Manipal, Karnataka, India | CGPA: 8.44 for 10

TECHNICAL SKILLS

- Programming Languages: C, C++, Python (NumPy, Pandas, Scikit-learn), Java, JavaScript, TypeScript, C#, SQL
- Web & Frontend: React, Angular, jQuery, Ajax, HTML, CSS, Tailwind CSS, Vite
- Backend & Frameworks: Node.js, Express, Flask, ASP.NET, MVC, Entity Framework, Linq, Socket.io, OpenAI API
- Architecture & APIs: Microservices, CQRS, Kafka, RESTful APIs, Swagger
- Cloud, DevOps & Security: Okta, Docker, GitHub Actions (CI), Azure DevOps, Postman, Azure CI/CD, AWS S3, Supabase, Vercel, Render
- Databases: PostgreSQL, MySQL, Stored Procedures, PIVOT
- Testing & Tools: MS Test, Unit Testing, Git, Jira, Agile/Scrum

PROJECTS

- **IPL Data Analysis:** Leveraged **PySpark** and **Databricks** on **AWS S3** datasets to predict match outcomes with **78% accuracy** using distributed data processing.
- **Network Traffic Analyzer:** Built a security tool in **Python (Scapy)** on **Linux** to perform **TCP/IP protocol analysis**, successfully identifying 5+ types of simulated network threats.

CampusBytes | Full Stack & GenAI Developer

- Architected a real-time social portal for **200+ users** using **React (Vite 7)** and **Node.js**, featuring **GenAI-powered** content generation via OpenAI API to increase user engagement by 25%.
- Optimized data synchronization using **Socket.io**, achieving sub-100ms latency for live post broadcasting, and integrated **Supabase (PostgreSQL)** for secure, domain-restricted authentication.
- Engineered a responsive "Windows Desktop" UI with Tailwind CSS v4, resulting in a **100/100 Lighthouse performance** score on Vercel deployments.

Cloud-based Storage Account | DevOps & Cloud Engineer

- Developed a **Docker-based** local emulation of **Azure Blob Storage**, improving local development data retrieval efficiency by **40%** and reducing cloud egress costs during testing.
- Automated container orchestration to simulate high-availability storage environments, streamlining the transition from local development to Azure production.

Computer-Aided Heart Sound Detection | ML & Health-Tech

- Developed a predictive health model using **Python and Signal Processing** to analyze heart sound frequency and intensity, achieving **85%+ accuracy** in identifying anomalies.
- Implemented feature extraction algorithms to forecast cardiac health trends, bridging the gap between raw biometric data and actionable medical insights.

ACTIVITIES

- Board Member of MIT Sports Club, MIT Cricket League, and MIT Basketball League of MIT Manipal.
- Organizer for HFS (Hospitality and Food Stalls) during Revels' 19 in College Fest.
- Volunteer for the sports category during Revels' 18 in College Fest.
- A member of the Clarivate's Bangalore Social Committee, which organizes events.

MERIT CERTIFICATE

Received an Award from the Associate Director and Senior Manager for the work I have delivered at Clarivate in 2022.